Movie Recommendation System:

1. Introduction

With the growing number of movies available across various platforms, users often find it difficult to choose what to watch. A recommendation system helps users discover movies based on their preferences.

This project focuses on building a simple movie recommendation system using Python that suggests movies based on genre and rating.

2. Problem Statement

Users face difficulty in selecting movies due to:

- Large number of available options
- Lack of personalized suggestions
- Time-consuming search process

The goal of this project is to develop a system that recommends movies efficiently based on user preferences.

3. Objectives

- To design a movie recommendation system
- To use a dataset for filtering and analysis
- To provide genre-based and rating-based suggestions
- To allow dynamic addition of new movies

4. Methodology

The project follows these steps:

1. **Data Collection**
    - A dataset of movies is stored in a CSV file
    - Fields include Title, Genre, and Rating
2. **User Interaction**
    - User selects options from menu

3. **Processing**
    - Filter movies by genre
    - Sort movies by rating
    - Filter based on minimum rating
4. **Output**
    - Display recommended movies

5. Tools and Technologies Used

- Programming Language: Python
- Data Storage: CSV file
- Platform: Local system / VS Code

6. Implementation

The system includes the following features:

- **Genre-based recommendation**
- **Top-rated movie display**
- **Filter by rating**
- **Add new movie to dataset**

The program reads data from a CSV file, processes user input, and displays results accordingly.

7. Challenges Faced

- Handling file operations correctly
- Managing user input errors
- Designing efficient filtering logic
- Debugging runtime errors

8. Results

The system successfully:

- Provides movie recommendations
- Allows users to add new movies
- Displays top-rated movies
- Improves user experience

9. Future Enhancements

- Add machine learning-based recommendations
- Develop a web-based interface
- Use APIs for real-time data
- Improve user interface

10. Conclusion

This project demonstrates how a simple recommendation system can help users choose movies efficiently. It also highlights the use of datasets and basic data processing techniques.

11. Learning Outcomes

- Understanding of recommendation systems
- Experience with Python programming
- File handling using CSV
- Data filtering and sorting